
Bayesian Optimization Hyperparameter Tuning Framework for Predictive Urban Gentrification Modeling

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Abstract

This paper presents a machine learning framework for predicting urban gentrification using spatial, demographic, and infrastructure-related datasets, using the City of Chicago as a case study. Motivated by the growing need for proactive anti-displacement policy in rapidly changing urban neighborhoods, the paper investigates whether early indicators of gentrification can be identified before large-scale resident displacement occurs. The proposed framework integrates numerous datasets from the United States Census Bureau and municipally tracked geospatial information between 2010 and 2020, from which a composite gentrification index serves as the model’s ground-truth classification target. The paper further explores the relative predictive value of different urban datasets and applies Bayesian optimization to improve machine learning hyperparameter selection and model performance. The framework aims to provide policymakers and urban planners with a data-driven tool capable of identifying neighborhoods at risk of future gentrification, enabling earlier intervention strategies to reduce displacement and preserve community stability.

1 Introduction

Can machine learning models be used to predict enigmatic socioeconomic phenomena? As data scientists discover the abilities of applied artificial intelligence upon various domains, a nascent field rich in data yet unexplored in implementation remains in the urban sphere. The world’s largest cities collect terabytes of information on everything—what gets built when, every event, every police incident, every car that parks on its streets. As a result, our cities are the perfect grounds from which to apply newly created algorithms upon vast datasets. This becomes especially important when considering the importance of positive outcomes within our municipalities. Cities shape lives—when done well, the communities within thrive.

The importance of community retention and promotion is especially relevant in a market of housing scarcity. In recent years, urban neighborhoods have seen a surge in gentrification—wealthier residents that move into less wealthy neighborhoods in search of cheaper housing and a better quality of life. These residents often attract other wealthy homebuyers to move in, raising median home prices and rents. This increase in property taxes and rents begins to drive out existing residents, fundamentally changing the character of the community. This paper stands to question: can we predict gentrification before it occurs, giving policymakers time to act to minimize displacement?

1.1 Literature Review

Gentrification is the process in which neighborhood investment increases property values and rents, resulting in displacement of existing residents for those of a higher income. Academia has long-debated the definition and metrics by which gentrification occurs, with no clear answer emerging [1]. One metric often debated by data scientists measures proximity from schools, hospitals, parks, and other amenities as a primary factor [1-4]. Eshtiyagh et al. (2023) proposed a graph-based multi-modal network, using proximity to schools, hospitals, and subway stations to predict gentrification in New York City, Chicago, and Los Angeles. Using a graph convolutional network, the authors used census data, building footprints from satellite imagery, and distance-based proximity as measures to predict gentrification using AUC as an evaluation metric [2]. Work by Reades et al. (2018) focuses more broadly upon socioeconomic characteristics, such as median house price, educational attainment of neighborhood residents, and percentage of workers by occupational class. By using random forests, the authors predict gentrification at a neighborhood scale in London [3]. A working paper by Jayne Yoo from the United States Census Bureau uses a variety of machine learning methods and datasets to identify features that predict gentrification across the Washington, DC metropolitan area. The author finds that out of the best performing method (Random Forest Classifier), features that are of greatest usefulness to predicting gentrification are income, age, distance to schools, and walkability, among others [5].

1.2 Motivation

A clear example of gentrification can be found in Chicago, Illinois. The neighborhood of Lincoln Park, located on the north side of the city near the waterfront, was historically a working-class neighborhood of factory workers—in the mid 20th century, the neighborhood held large Hispanic, particularly Puerto Rican, and Black communities. However, in the 1970s, a large number of affluent white residents began moving in, renovating the neighborhood and driving up property values [6]. In 1970, the median house value in Lincoln Park was \$109,590. By 1980, it had more than doubled to \$262,704. By 1990, it more than doubled again to \$540,395. Such dramatic increases in property values caused significant increases in rent, and causing less wealthy residents to move away. By the 1990 census, Lincoln Park was 85% white, an 8 point increase from 1980. The percentages of Black and Latino residents had decreased by around 8 percentage points in the same time frame [7].

Cities have policies that can be taken to prevent gentrification-induced displacement, such as what former residents of Lincoln Park experienced, within their neighborhoods. However, policymakers may only act upon these policies through active mitigation or proactive planning rather than preemptive action. Through machine learning, if early-stage gentrification can be predicted, policymakers can minimize resident displacement through highly targeted municipal action. This project aims to answer if distance-based amenity metrics can provide an effective surrogate to predict early-stage gentrification, in which prior trends obtained via distance metrics can be extrapolated as effective predictors.

2 Methodology

This model consists of multiple phases in its construction, unique to the city being analyzed in order to prevent out-of-distribution data extrapolation. First, the ground-truth for what has been considered gentrified in the past will be obtained. Varying geospatial datasets of the municipality being studied are run through a gravity model, comparing their distances to census tracts, and comparing the gravity scores to ground-truth information on past-gentrification data to analyze each dataset’s significance to gentrification. Once the statistically insignificant datasets are filtered, the remaining datasets are used to hyperparameter tune a model using Bayesian optimization. The best-tuned model is used to predict gentrification on unseen data.

The city of Chicago is used as a case study for this model. This was done in part due to the high availability of data, relatively unified dataset, and lack of natural barriers which might otherwise inhibit Euclidean distance calculations. Regions such as Los Angeles consist of many small, poly-centric municipalities that may vary in data collection methods. Likewise, New York City’s large natural barriers such as Jamaica Bay and New York Harbor prevent usage of cross-borough Euclidean distance measurements.

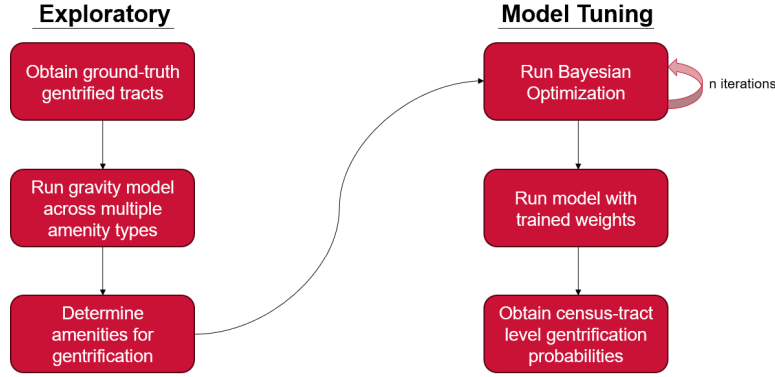


Figure 1: Bayesian model pipeline, divided into steps.

2.1 Data Pre-Processing

Several datasets were gathered across the United States Census Bureau and City of Chicago’s Data Portal:

- US Census LODES data, which consists of census-block granularity job employment and location data at Census years 2010 and 2020.
- Chicago Park District Facilities, which contains the location data of all park facilities. Larger parks with more facilities will have more location points and thus greater weight.
- CTA ‘L’ Stops, which contains the location data for all high-frequency rail stations.
- US Census Demographics Data, which consists of median gross rent, median household income, percentage of population in poverty, median gross rent, median home value, and number of individuals with a college degree or higher (see Appendix A), at a census-tract level resolution at Census years 2010 and 2020.
- Chicago Police Department Crime Data, which lists every reported crime between 2001 and present-day, the date of occurrence, and the coordinates at which the crime occurred.
- Chicago Building Permits, which lists every building permit filed between 2006 and present-day, the date of occurrence, and the coordinates of the building site.

To simplify the temporal changes in demographics data between Census years 2010 and 2020, the two Census demographics datasets are merged to display the differences in their values. For example, if within a census tract the number of college educated workers is 20 in 2010 and 50 in 2020, then the differential dataset displays the number of college educated workers as 30. Because Census tracts change shape between census years, an additional US Census Tract Relationship Boundary data set enables ease of comparison between the two.

For ease of processing during model training, the datasets obtained were standardized to consistent formats. Because Census data is specified at the Census tract and block resolution, each census tract and block listed in the dataset was given a corresponding latitude and longitude coordinates associated with the unit’s centroid for ease of comparison with datasets identified by coordinate points. Because my lowest dataset resolution is at the census-tract level, this model only measures gentrification at that resolution, despite higher resolution datasets (census-block level, latitude/longitude) used in this project.

Building permit and crime data is filtered and divided into two separate datasets. In accordance with Census data intervals, crime and building permits are split across calendar years 2006-2010 and 2016-2020.

2.2 Exploratory Phase

The purpose of this phase is to determine which datasets have a measurable impact upon the prevalence of gentrification.

For my ground-truth, I used US Census data to identify the top $k\%$ of Census tracts with greatest changes in socioeconomic indicators. The differential census data is converted into a z-score. Each census tract's "gentrification score" is given as a weighted sum of z-scores:

$$g_i = \sum \frac{x_i - \mu}{\sigma}$$

The top $k\%$ of Census tracts by gentrification score are labeled as having gentrified between 2010 and 2020. The value of k within the Chicago data was set to 10% based on manual analysis of neighborhood change in high-scoring Census tracts. The limitations of this ground truth labeling method will be discussed later in Section 4.1.

A distance-based gravity heuristic is utilized to calculate the distance between each census tract and a particular dataset of amenities. For a given census tract, per amenity $j \in A$, where j is a data point within dataset A , within fixed radius r obtain the z-score of the sum of the amenity category, multiplying by the inverse distance from the parcel.

$$G_i = z \left(\sum_{j \in A: d_{ij} < r} \frac{1}{d_{ij} + \epsilon} \right)$$

Our gravity weights are then placed into a Bayesian logistic regression model to estimate the probability that a census tract is classified as gentrifying. Let $Y_i \in \{0, 1\}$ denote the observed gentrification label for tract i , where $Y_i = 1$ indicates a gentrifying tract and $Y_i = 0$ indicates a non-gentrifying tract. The model assumes that each outcome follows a Bernoulli distribution, $Y_i \sim \text{Bernoulli}(p_i)$, where p_i is the probability that tract i is gentrifying. The probability is modeled through a logistic link function:

$$p_i = \sigma(\eta_i) = \frac{1}{1 + e^{-\eta_i}}$$

Where the linear predictor is defined as

$$\eta_i = \alpha + x_i^\top \beta$$

Where α is the intercept and $\beta_1, \dots, \beta_k$ are the predictors. Posterior sampling is then used to estimate the distribution of α and β conditional upon the ground-truth data. The resulting posterior samples indicate the distribution and uncertainty of the amenity towards predicting gentrification. Within our Chicago datasets, only building permits and job data (divided by income levels) presented a statistically significant positive distribution towards gentrification. The implications of this relative to other academic work on predictive gentrification will be discussed further in Section 4.2.

2.3 Model Tuning

From our datasets that are statistically significant towards gentrification, we construct our predictive model. We update our per-tract gravity equation with 2 hyperparameters for each amenity to include a weight and a radius:

$$G_i(r, \alpha) = z \left(\sum_{j \in A: d_{ij} < r_A} \frac{\alpha_A}{d_{ij} + \epsilon} \right)$$

The predictive component of the model converts the standardized values into probabilities using logistic regression. Per tract i , the model estimates the probability of gentrification as

$$P(Y_i = 1) = \frac{1}{1 + e^{-(\beta_0 + z_i^\top \beta)}}$$

where z_i is the vector of standardized gravity-features, and β contains the feature coefficients, with β_0 as the intercept. The data is split into training and validation sets, and the model is evaluated using

negative log loss:

$$-L = - \left[-\frac{1}{n} \sum_{i=1}^n (Y_i \log(p_i) + (1 - Y_i) \log(1 - p_i)) \right]$$

This is done such that higher values correspond to better predictive performance. Class balancing is utilized as only the top $k\%$ of tracts are labeled as gentrifying, making the positive class rarer than the negative.

Bayesian optimization is then used to tune the hyperparameters $\alpha_0, \dots, \alpha_k, r_0, \dots, r_k$. Thompson sampling is utilized to choose new candidate hyperparameters likely to improve validation performance. The optimization uses Sobol-sampled initial points, repeatedly fitting the Gaussian process to observed $(x, f(x))$ pairs, sampled, and then evaluated, and compared against the best performing vector x^* . This process is repeated for n iterations.

In my Chicago case study, the optimized gravity parameters x^* , consists of 8 hyperparameters:

$$\alpha_{building_permits}, \alpha_{low_income_jobs}, \alpha_{medium_income_jobs}, \alpha_{high_income_jobs}, \\ r_{building_permits}, r_{low_income_jobs}, r_{medium_income_jobs}, r_{high_income_jobs}$$

These have been tuned on data at time t_0 (2010), in order to predict gentrification at t_1 (2020). The gravity model is run once more, but with parameters x^* on data at time t_1 (2020) to predict gentrification at t_2 (2030) upon Census tracts that were not already classified as gentrifying at t_0 . The resulting prediction for each tract becomes

$$\hat{p}_{i,t_2} = P(Y_{i,t_2} = 1 | z_{i,t_1}, x^*)$$

representing the estimated probability that tract i will be undergoing early-stage gentrification by t_2 .

3 Results

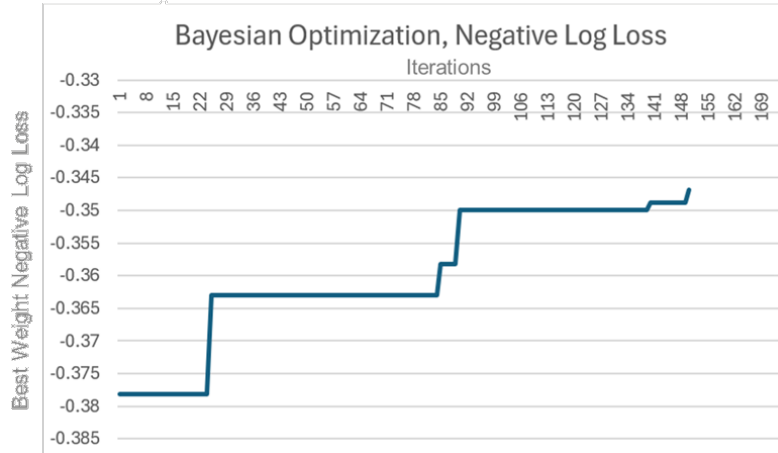


Figure 2: Best Negative Log Loss by iteration.

Figure 2 shows the overall performance of the model over 150 iterations. Note that there are many periods in which the graph appears flat, as the iterations that the sampler generates are worse performing than the best vector x^* . This will be discussed further in Section 4.4.

Figure 3 shows the predictions made by the model at t_2 (2030). The darker the census tract, the greater probability that the tract will undergo gentrification by t_2 . The lighter the census tract, the lower the probability. Note that any Census tracts above the city's median rent or income are excluded from the dataset, as including already wealthy tracts skews the data towards areas already considered gentrified. Accuracy of these predictions will be discussed further in Section 4.

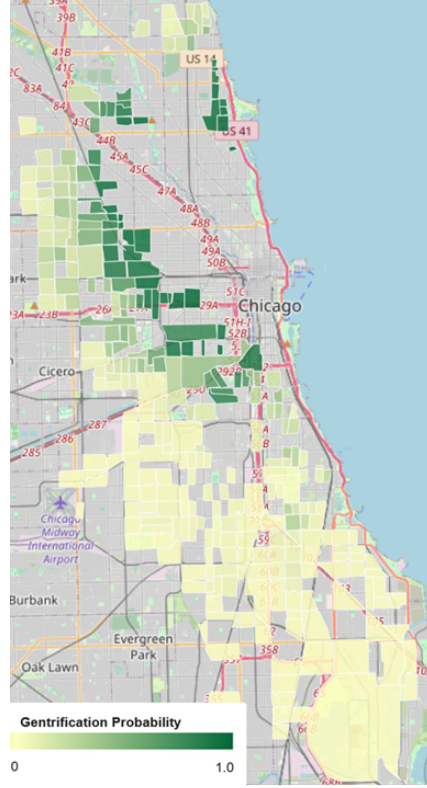


Figure 3: Gentrification probabilities at t_2 (2030), visualized by census tract.

4 Discussion

There exist many limitations to the gentrification system developed here. Among them are broad issues of unclear ground truth and conflicting definitions of gentrification which make such a system difficult to evaluate beyond computational measures of distribution prediction.

4.1 Ground Truth

The ground truth used to classify Census tracts between t_0 and t_1 is imperfect. While income and education are good indicators at determining the kinds of people who might be so-called "gentrifiers", they do not fully encompass the definition of increased price-induced displacement. The change in demographics data was initially considered as a metric within the dataset, however, was excluded in the final model. Changing demographics can reflect a wide range of scenarios, from immigrant communities that move in to additional residents that reflect new housing construction, not necessarily displacement as a result of elevated property values. Within the methodology of Eshtiyagh et al., the ground truth used for prior gentrification comes from the Urban Displacement Project, a resource at the University of California Berkeley that aims to understand and catalog gentrification as it occurs in urban areas, with visualizations that display their methodology's results (see Figure 4) [8]. However, this creates new issues by offloading ground truth to an external source. As gentrification does not have a singular academic definition, many labels used by the Urban Displacement Project does not have consensus among academics. This will be further discussed in Section 4.3.

4.2 Amenity Datasets

The datasets evaluated for statistical significance in this model were arbitrary, potentially missing many factors that contribute towards gentrification-induced investment. Datasets such as parks, rail stations, and jobs were selected due to their ease of availability and static availability across time. Additional datasets such as hospital locations are more difficult to locate, perhaps requiring external

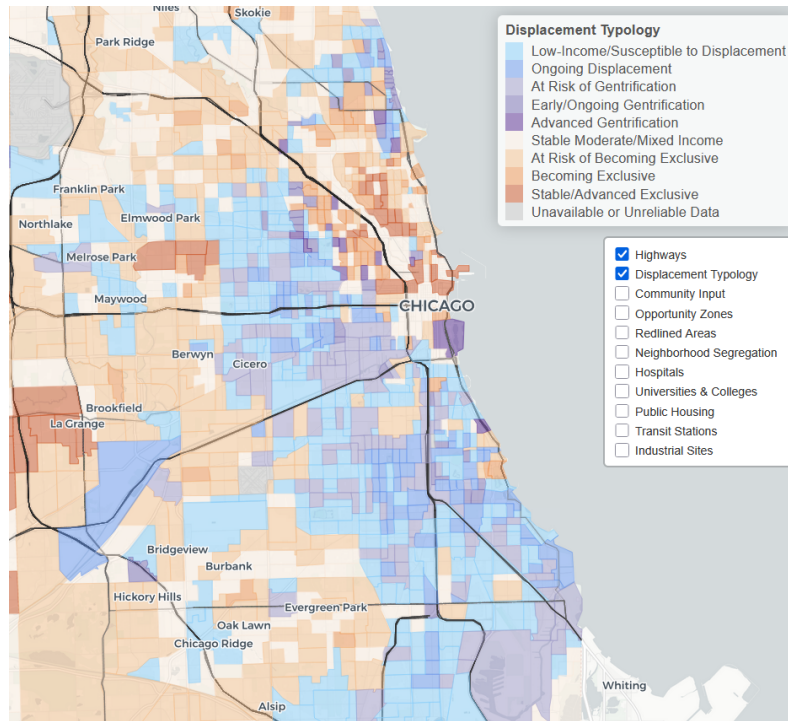


Figure 4: Gentrification categories by Census tract, as defined by the Urban Displacement Project.

datasets from tools such as OpenStreetMap to obtain. Furthermore, I found that many datasets found by other academic works to have statistical significance did not have such significance within my model. Work by Rigolon and Németh, and statistically validated by Eshtiyagh et al. conclude that access to essential facilities such as schools, hospitals, and subway stations showed the strongest correlations with gentrification [2,4]. However, in my statistical analysis, this proved not to be the case. I hypothesize that while such conclusions may prove valid within other cities— Eshtiyagh et al. evaluated New York City and Los Angeles in addition to Chicago— such conclusions are not valid when merely looking from the perspective of Chicago itself. Perhaps this is because Chicago’s sprawling subway system covers many quiet, low density residential areas far from the city’s core, rather than merely dense commercial streets. By the theory of amenity gravity, such areas are unlikely to gentrify due to their far distance from the city’s dense core of shops, restaurants, and jobs. A similar explanation would appear as to why parks were not statistically significant towards gentrification in both my own and the analysis of Eshtiyagh et al. Further testing among a variety of cities in the United States will be necessary to determine the validity of this theory.

4.3 Defining Gentrification

A consistent point of contention throughout this project has arisen from the definition of the problem itself: what is gentrification and how do you measure it? Work by Jayne Yoo for the US Census Bureau cites the Department of Housing and Urban Development, defining "a form of neighborhood change that occurs when high-income groups move to low-income areas, potentially altering the cultural and financial landscape of the original neighborhood" [5]. Eshtiyagh et al. cites the initial definition coined by Ruth Glass, defined as "all or most of the working class occupiers [to be] displaced and the whole social character of the district [to be] changed" [2]. Urban planning scholars disagree on key details: A literary analysis by Nicholas Finio finds that while they generally agree that gentrification is the result of an influx of investment and higher income, higher education residents moving into a neighborhood, they disagree on tangential outcomes and other variables associated with the phenomena. Do racial categories have a factor when considering when an area is to be considered gentrifying or gentrified, such as wealthier white residents moving into a majority-black neighborhood? What other variables, such as the median age of residents, housing stock, or other complex conditions should be considered? Finio concludes that there is very little agreement on

the complex processes used to define the phenomena beyond the basic definitions of income and education. [1]

These overlapping and occasionally conflicting definitions collide upon evaluation of model predictions. An assessment of Figure 3 reveals that the areas with the greatest chance of gentrification are directly north and west of downtown, adjacent to neighborhoods above the city's median rent or income, with low probabilities of gentrification occurring elsewhere. The Urban Displacement Project seems to agree, with the neighborhood known as "Uptown" in the North listed as ranging between "Early/Ongoing Gentrification" and Advanced Gentrification [8]. A check of the US Census Bureau American Community Survey's average rent prices between 2019 and 2024 finds that the median rent within uptown Census tracts increased by approximately \$400. There seems to be solid agreement that Uptown is being gentrified.

Other neighborhoods yield more mixed results. Chicago's Chinatown, located just south of downtown, is marked with a high probability of gentrification by 2030 according to my model. Local news seem to agree, with recent news articles discussing rising rents, with local residents raising concerns that "they've already seen their property tax go up so much...They feel the gentrification pressure. 'Can I stay in Bridgeport? Can I stay in Chinatown?'" [9]. Despite this evidence, the Urban Displacement Project categorizes the Chinatown neighborhood among their lowest risk of gentrification, known as "Low-Income/Susceptible to Displacement" [8]. Woodlawn, a neighborhood located just south of the University of Chicago, also receives mixed signals. The Urban Displacement Project classifies the neighborhood as "Early/Ongoing Gentrification", and local news articles suggest similar themes, as the area experiences investment from the construction of the Obama Presidential Center nearby [10]. ACS data suggests similarly, with median rent increasing by approximately \$400 between 2019 and 2024. However, the model gives the whole neighborhood a 2.4% chance of gentrification by 2030. This suggests a number of things: because gentrification is a spectrum rather than a binary classification, my system may ignore early signs of gentrification, such as large-scale outside investment and development. Because the Obama Presidential Center is a single building rather than many, and because my model does not distinguish between permit types, the construction of a single building may appear to be of non-importance to the model. A third explanation might be that, because of the many definitions of gentrification, in its current state, many definitions conflict as to if the process is even taking place in Woodlawn.

4.4 Model Performance

Another point of contention that remains is the performance by which the Bayesian model performed hyperparameter tuning. Analyzing Figure 2, there are few points at which the model significantly improves, with the best performing hyperparameters remaining constant throughout most iterations. In the future, I may instead consider using other forms of sampling to address this shortfall such as Expected Improvement or Upper Confidence Bound, while also using AUC as an additional performance metric from which to gauge the model's improvement over time.

4.5 Future Work

Any further work on this project must address the limitations mentioned in Section 4. Additional work may include redefining ground truth to match various definitions as proposed by academics, such as the Urban Displacement Project. However, as ground truth continues to be elusive in our understanding of gentrification, I propose a sentiment analysis model of gentrification, aided by definitive measures of the subject such as changes in rent prices in low-income neighborhoods and percentage of individuals with college degrees. Such a model would inspect articles on neighborhoods within a given city, weighted by the city's changes in their socioeconomic conditions over time. Such a model may provide a better ground truth for future gentrification prediction models.

As amenity categories studied remain arbitrary, I propose an amenity-space autoencoder neural network as a measure to determine datasets to collect. Such a system might inspect geospatial data from OpenStreetMap across a whole city, determining the categories of urban form and their relations to each other to form an urban latent space. Such a latent space, when combined with ground truth on prior gentrification, may provide valuable insight towards what causes an area to gentrify beyond generalized theories of distance to amenities.

5 Conclusion

Gentrification as a generalized problem provides an interesting metric from which large-scale urban data can be analyzed through the work of machine learning models. Using Bayesian Optimization, I created a model that can hyperparameter tune within 150 iterations to make predictions on near-term early-stage gentrification, teaching me both fundamentals of Bayesian optimization usage using BoTorch while exploring varied datasets to understand complex phenomena. While such a model offers much promise for city planners in their efforts to ensure equitable access to housing for all, lingering questions around the problem's boundaries pervade the topic. This ultimately creates predictions that remain difficult to interpret and provide useful action upon, and require additional research to draw complete conclusions upon the validity of model outputs.

Acknowledgements

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A Census Dataset Variables

The data collected from the 2010 and 2020 US Census covers the following variables:

- B17021_001E: POVERTY STATUS OF INDIVIDUALS IN THE PAST 12 MONTHS BY LIVING ARRANGEMENT
- B19013_001E: MEDIAN HOUSEHOLD INCOME IN THE PAST 12 MONTHS (IN 2010 INFLATION-ADJUSTED DOLLARS)
- B25064_001E: MEDIAN GROSS RENT (DOLLARS)
- B25077_001E: MEDIAN VALUE (DOLLARS)

The following subcategories of B15002: SEX BY EDUCATIONAL ATTAINMENT FOR THE POPULATION 25 YEARS AND OVER were aggregated to obtain a more generalized educational variable:

- B15002_015E: Male, Bachelor's degree
- B15002_016E: Male, Master's degree
- B15002_017E: Male, Professional school degree
- B15002_018E: Male, Doctorate degree
- B15002_032E: Female, Bachelor's degree
- B15002_033E: Female, Master's degree
- B15002_034E: Female, Professional school degree
- B15002_035E: Female, Doctorate degree